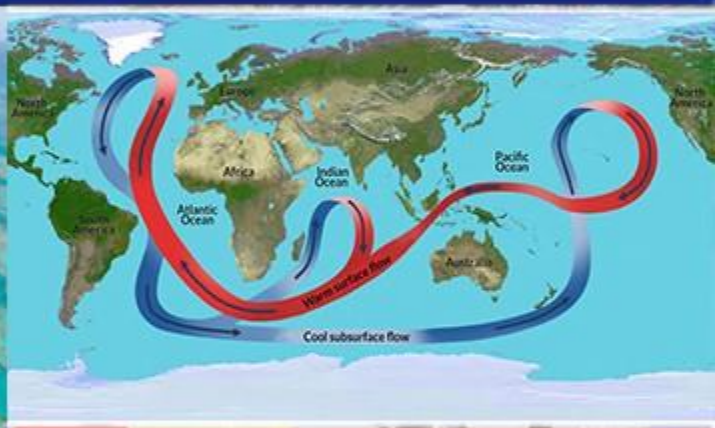


GEOGRAPHY

PHYSICAL GEOGRAPHY



OCEANOGRAPHY

MODULE - 2

TEMPERATURE

TEMPERATURE OF OCEAN WATERS



- 1 This section deals with the spatial and vertical variations of temperature in various oceans.
- 2 Oceans water get heated up by the solar energy just as land.
- 3 However, the process of heating and cooling of the oceanic water is slower than land.

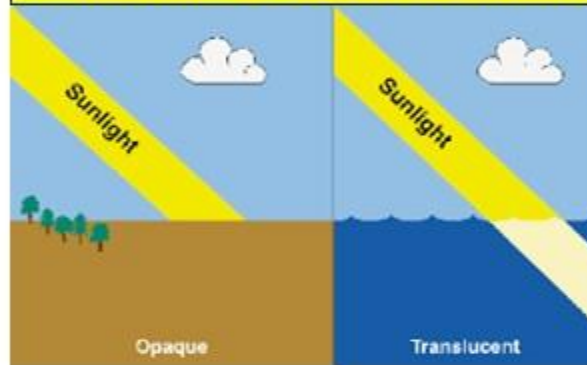
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FACTORS AFFECTING TEMPERATURE DISTRIBUTION

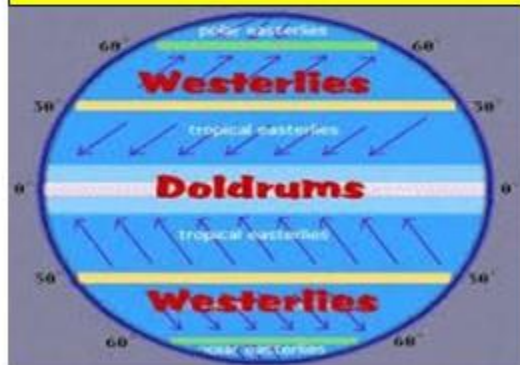
1. LATITUDES



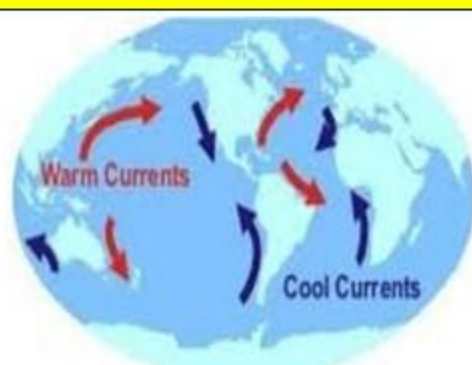
2. NATURE OF LAND & WATER



3. PREVAILING WINDS



4. OCEAN CURRENTS



The factors which affect the distribution of temperature of ocean water are :

1. Latitude
2. Unequal distribution of land and water
3. Prevailing Wind
4. Ocean Currents
5. Minor factors will include local weather conditions like storms, cyclones, fog, cloudiness; shape of the seas.

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FACTORS AFFECTING TEMPERATURE DISTRIBUTION

1

LATITUDE

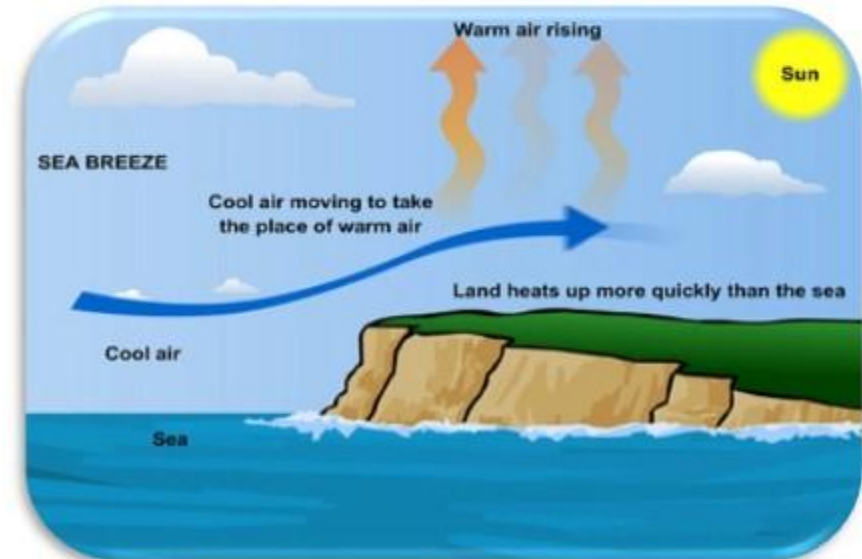
The temperature of surface water **decreases from the equator towards the poles** because the amount of insolation decreases poleward.



2

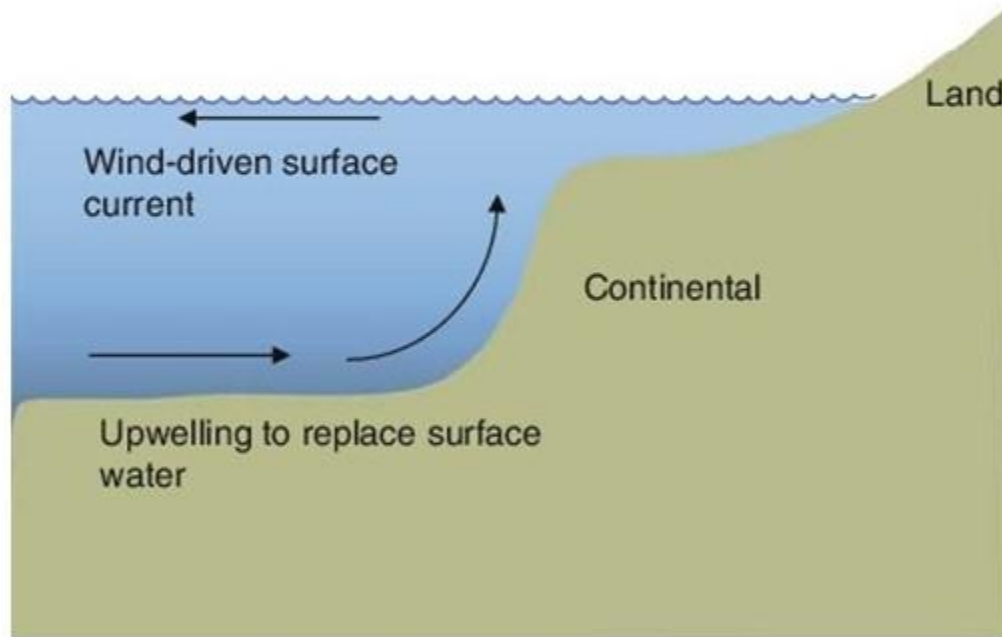
UNEQUAL DISTRIBUTION OF LAND AND WATER

The oceans in the **northern hemisphere** receive more heat due to their contact with **larger extent of land** than the oceans in the southern hemisphere.



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FACTORS AFFECTING TEMPERATURE DISTRIBUTION



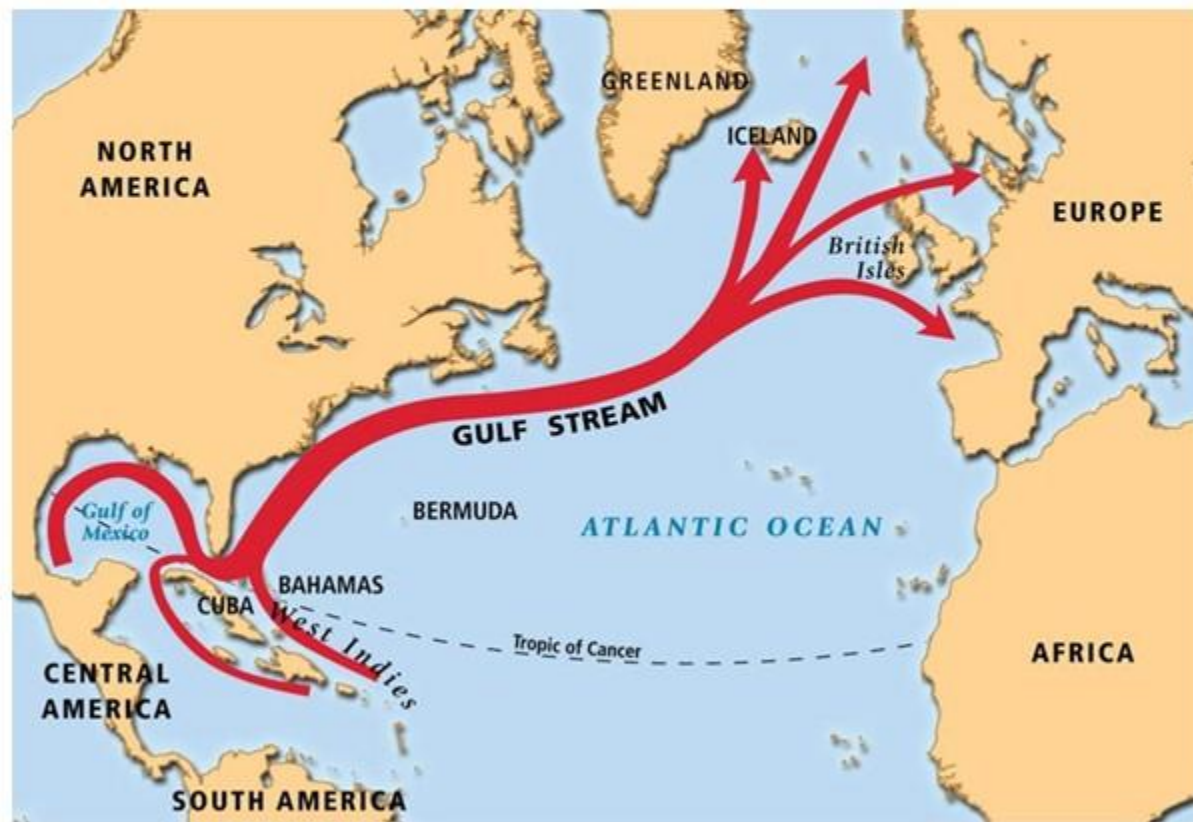
Upwelling of cold water as the warm surface water is being driven away

3 PREVAILING WIND

- The winds blowing from the land towards the oceans **drive warm surface water away from the coast** resulting in the **upwelling of cold water from below**.
- Thus, the replacement of warm water by cold water results into the **longitudinal variation in the temperature in oceans**.

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FACTORS AFFECTING TEMPERATURE DISTRIBUTION



Warm Gulf Current raises the temperature of eastern coast of North America

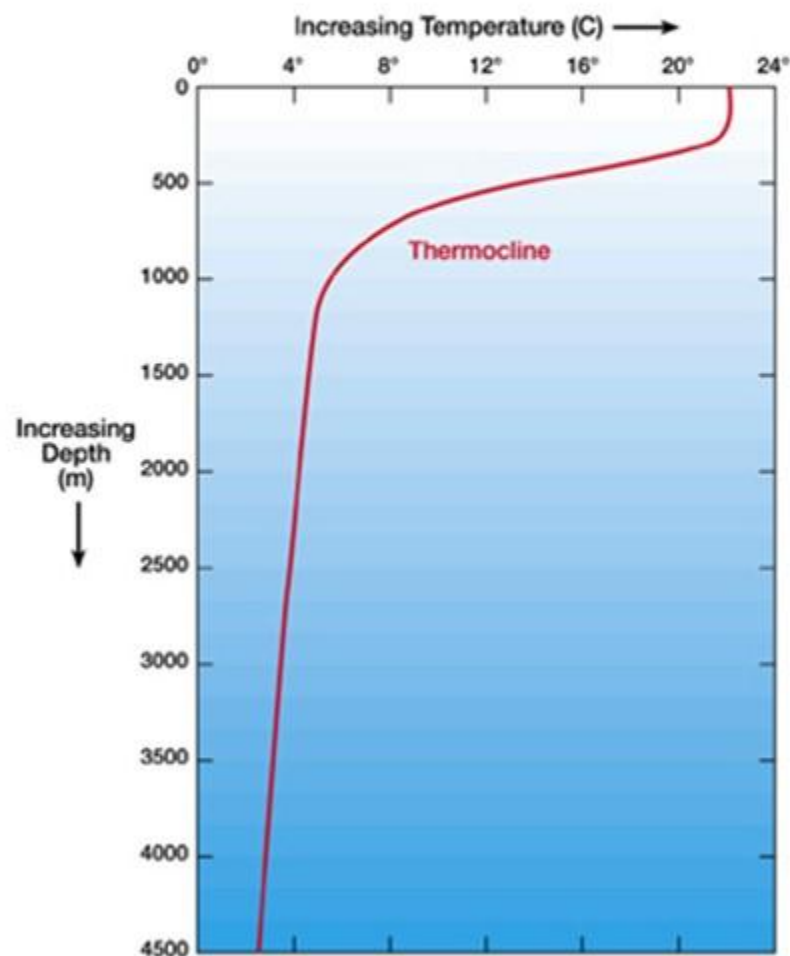
4

OCEAN CURRENTS

- Warm ocean currents raise the temperature in cold areas while the cold currents decrease the temperature in warm ocean areas.
- Gulf stream** (warm current) raises the temperature near the eastern coast of North America and the West Coast of Europe while the **Labrador current** (cold current) lowers the temperature near the north-east coast of North America.

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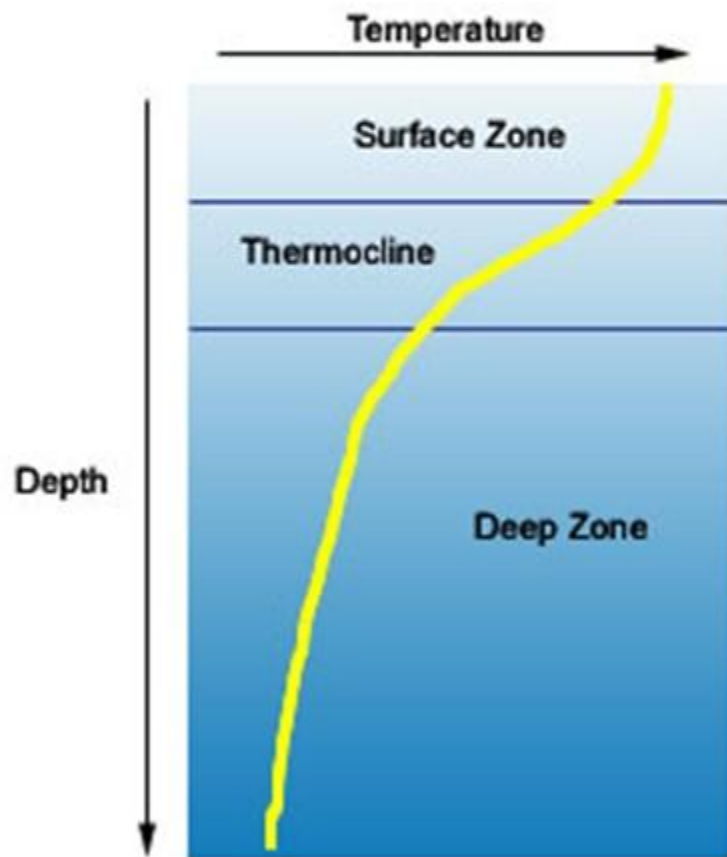
HORIZONTAL AND VERTICAL DISTRIBUTION OF TEMPERATURE



- 1 The temperature-depth profile for the ocean water shows how the temperature decreases with the increasing depth.
- 2 The profile shows a boundary region between the surface waters of the ocean and the deeper layers.
- 3 The boundary usually begins around 100 – 400 m below the sea surface and extends hundreds of metres downward.
- 4 This boundary region, from where there is a rapid decrease of temperature, is called the thermocline.

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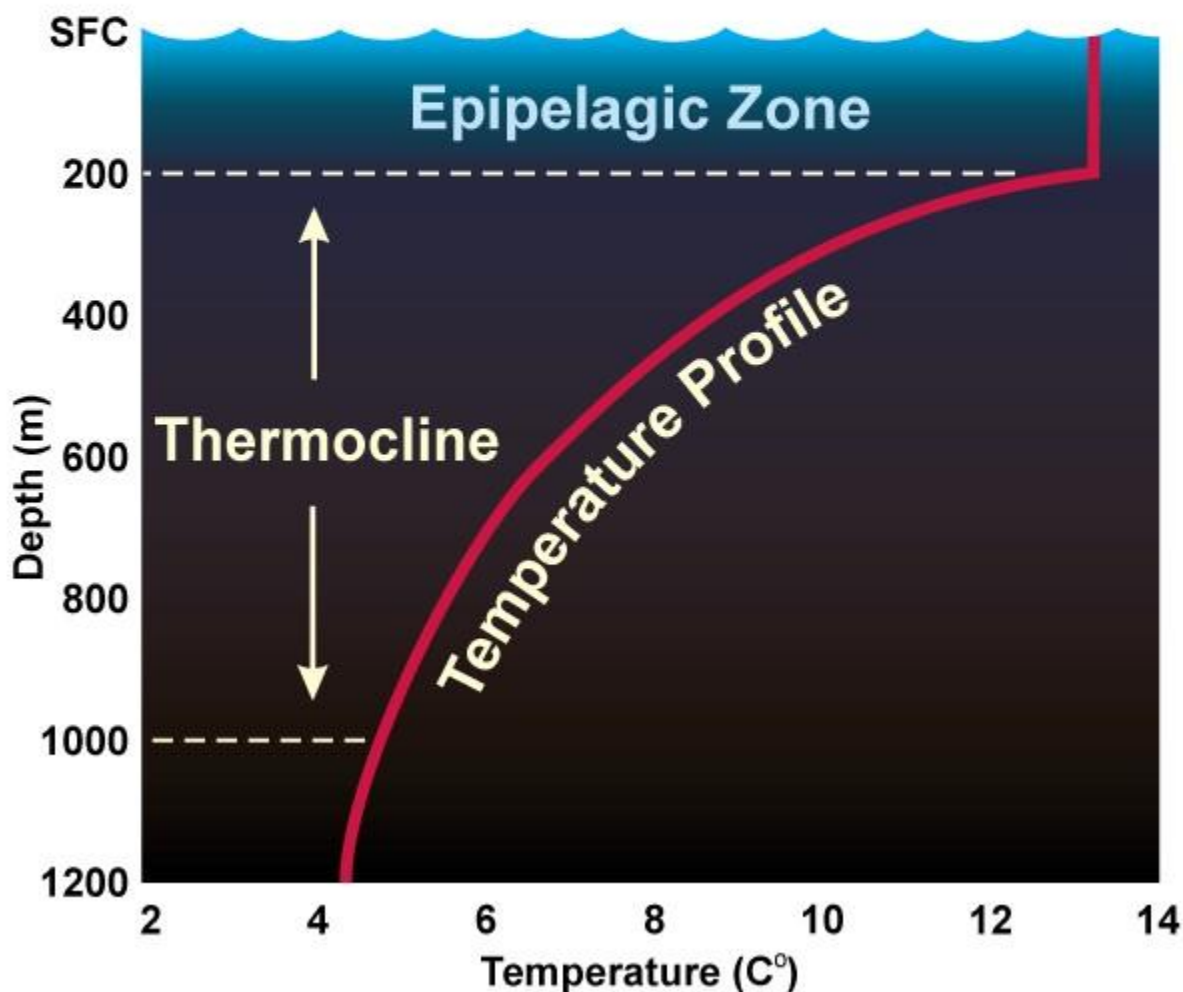
HORIZONTAL AND VERTICAL DISTRIBUTION OF TEMPERATURE



- 5 About 90 per cent of the total volume of water is found below the thermocline in the deep ocean.
- 6 The temperature structure of oceans over middle and low latitudes can be described as a **three-layer system** from surface to the bottom.
- 7 The **first layer** represents the top layer of warm oceanic water and it is **about 500m thick** with temperatures ranging **between 20° and 25°C**.

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HORIZONTAL AND VERTICAL DISTRIBUTION OF TEMPERATURE



8 This layer, within the tropical region, is present throughout the year but in mid latitudes it develops only during summer.

9 The **second layer** called the **thermocline** layer lies below the first layer and is characterised by **rapid decrease in temperature** with increasing depth.

10 The thermocline is **500 – 1,000 m** thick.

11 The **third layer** is **very cold** and extends up to the deep ocean floor.

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HORIZONTAL AND VERTICAL DISTRIBUTION OF TEMPERATURE



12 In the Arctic and Antarctic circles, the surface water temperatures are close to 0°C and so the temperature change with the depth is very slight.

13 Here, only one layer of cold water exists, which extends from surface to deep ocean floor.

14 The average temperature of surface water of the oceans is about 27°C and it gradually decreases from the equator towards the poles.

A world map illustrating the average surface water temperature by latitude. The map uses a color scale from blue (colder) to red (warmer). Key latitude lines are marked with dashed lines and corresponding temperature labels in colored boxes:

- 0°C Near Poles:** Indicated by yellow boxes at the top and bottom of the map.
- 14°C at 40° Latitude:** Indicated by blue boxes at approximately 40° North and 40° South.
- 22°C at 20° Latitude:** Indicated by red boxes at approximately 20° North and 20° South.
- Average temperature of surface water is about 27°C:** Indicated by a black box along the equator (0° latitude).

The map also labels major oceans (Arctic, North Atlantic, South Atlantic, Indian, North Pacific, South Pacific), seas (Greenland, Barents, Caribbean, Gulf of Mexico, Red Sea, Persian Gulf, Arabian Sea, Bay of Bengal, Andaman Sea, Celebes Sea, Philippine Sea, Japan Sea, Korea Bay, Yellow Sea, Bohai Sea, East China Sea, South China Sea, Taiwan Strait, Luzon Strait, Mindanao Strait, Sulu Sea, Celebes Sea, Molucca Sea, Timor Sea, Tasman Sea, Coral Sea, Ross Sea, Weddell Sea, Antarctic Sea), and major landmasses (North America, South America, Europe, Africa, Asia, Australia, Antarctica, Greenland, Iceland, British Isles, Scandinavia, Alps, Pyrenees, Iberian Peninsula, Italian Peninsula, Balkans, Greece, Turkey, Middle East, Central Asia, South Asia, Southeast Asia, East Asia, Oceania).

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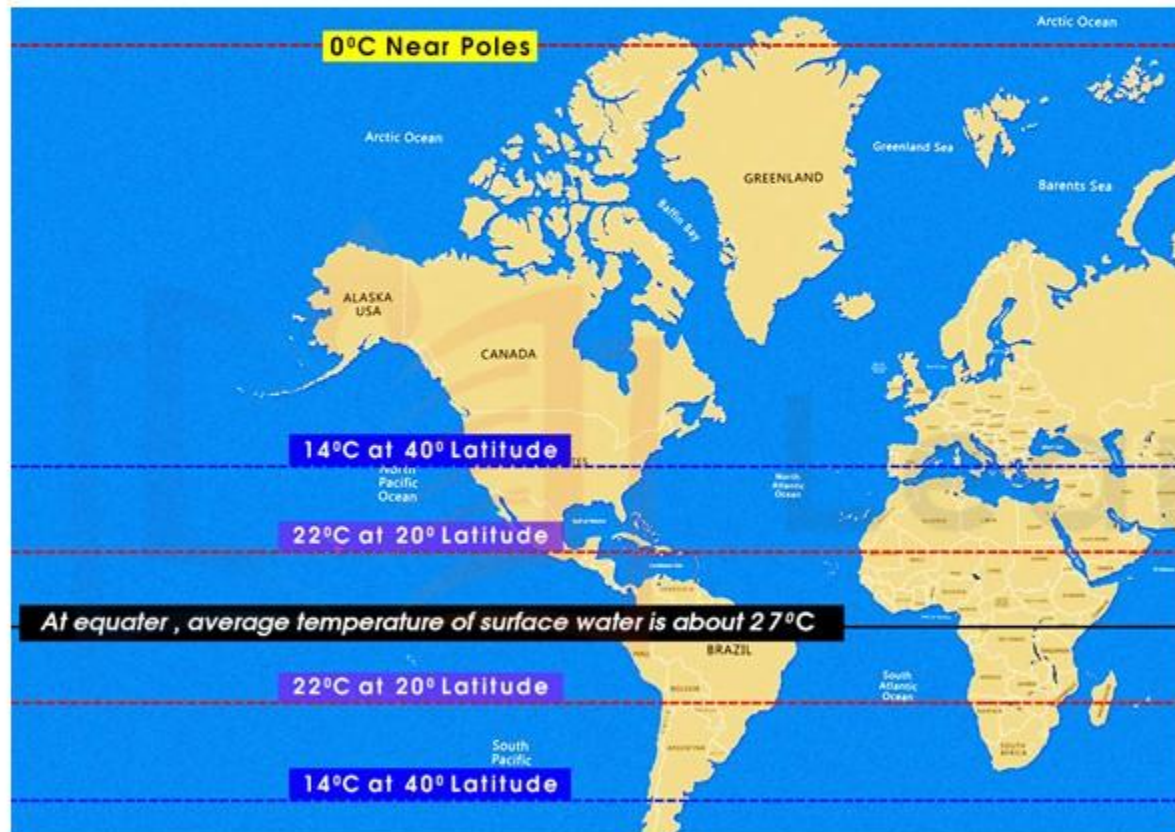
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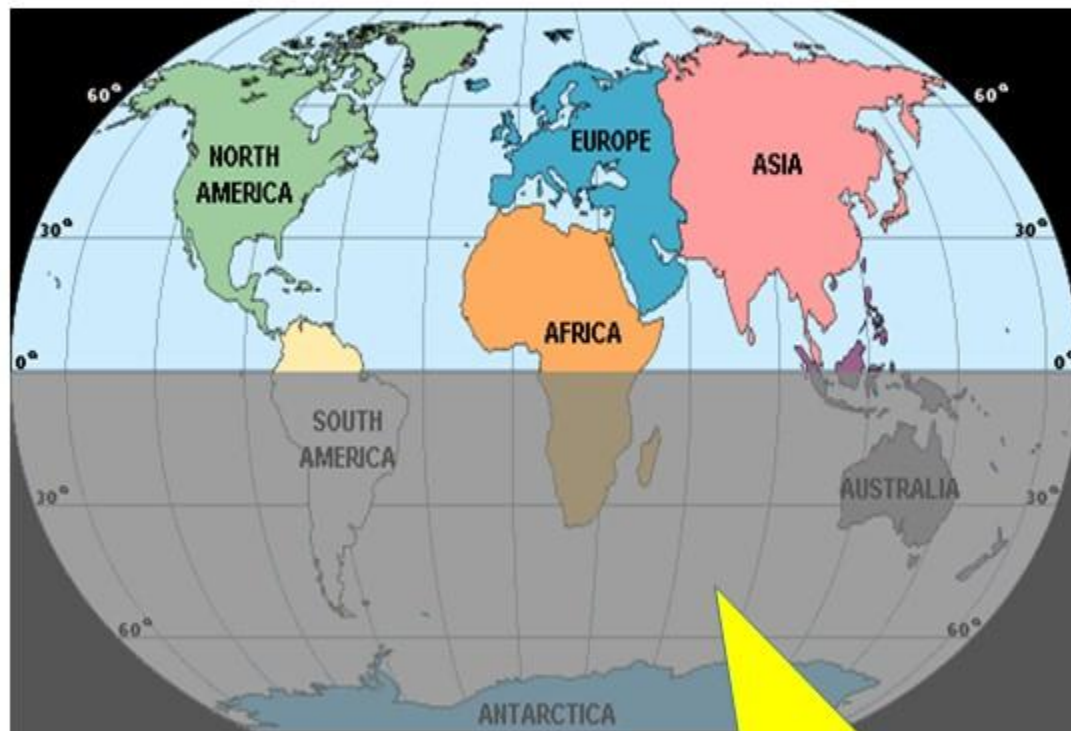
HORIZONTAL AND VERTICAL DISTRIBUTION OF TEMPERATURE



- 15** The rate of decrease of temperature with increasing latitude is **generally 0.5°C per latitude**.
- 16** The average temperature is around **22°C at 20° latitudes, 14° C at 40° latitudes and 0° C near poles**.
- 17** The oceans in the northern hemisphere record relatively higher temperature than in the southern hemisphere.

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HORIZONTAL AND VERTICAL DISTRIBUTION OF TEMPERATURE



Southern Hemisphere dominated by oceans

- 18 The highest temperature is not recorded at the equator but slightly towards north of it.
- 19 The average annual temperatures for the northern and southern hemisphere are around 19°C and 16°C respectively.
- 20 This variation is due to the unequal distribution of land and water in the northern and southern hemispheres.

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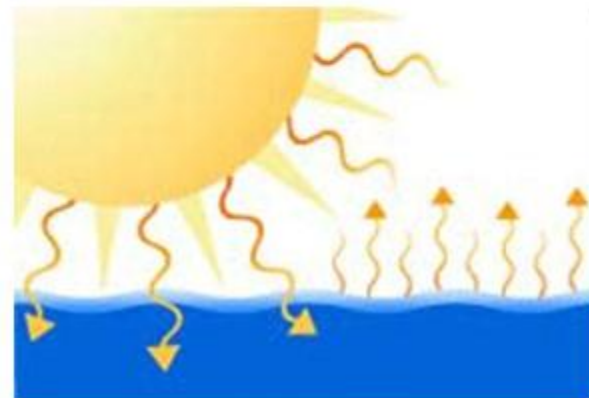
HORIZONTAL AND VERTICAL DISTRIBUTION OF TEMPERATURE

- 21** Generally, the maximum temperature of the oceans is at their surfaces.



- 22** This is because they directly receive the heat from the sun and the heat is transmitted to the lower sections of the oceans through the process of convection.

- 23** It results into decrease of temperature with the increasing depth, but the rate of decrease is not uniform throughout.



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